

Lecture 1: Integrability and Thermodynamic Bethe Ansatz in Quantum Field Theory

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You can find these lecture notes and some exercises at:
<https://olallacastroalvaredo.weebly.com/pentegost-school-25.html>



These $(1+1)$ -dimensional models

I got really fascinated by these $(1 + 1)$ - dimensional models that are solved by the Bethe ansatz and how mysteriously they jump out at you and work and you don't know why. I am trying to understand all this better.

R. Feymann, Asian-Pacific Physics News, Vol. 3, 22 (1988)

What is Integrability in Physics?

- There are different definitions but mostly it is a combination of the following three features: **Lioville Integrability**, **Conserved Quantities** and **Exact Solvability**.
- Classical Mechanics: The **two-body problem in Newtonian gravity** has sufficient symmetries and conserved quantities to allow for exact solutions.
- Quantum Mechanics: the **harmonic oscillator** and the **hydrogen atom** are integrable systems due to their symmetries and conserved quantities.
- Nonlinear Systems: The **Korteweg–de Vries (KdV) equation**, which describes waves on shallow water surfaces has an infinite number of conserved quantities and exhibits soliton solutions, which are stable, localized, shape-preserving waves.

Integrability in Quantum Field Theory

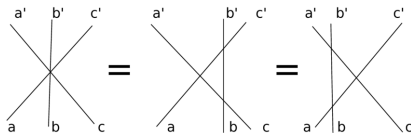
- In 1+1D there is also a notion of integrability in QFT. The consequences of integrability in this case are also very strong: **exact, non-perturbative, S -matrices, form factors, and correlation functions.**
- All scattering processes can be factorised into two-body scattering processes and there is no particle production.

$$S_{ab}^{cd}(\theta_a - \theta_b) = \text{starburst} = {}_{\text{out}}\langle Z_c(\theta_a)Z_d(\theta_b) | Z_a(\theta_a)Z_b(\theta_b) | 0 \rangle_{\text{in}}$$

- All the scattering properties of the theory are then encapsulated into one single object: $S_{ab}^{cd}(\theta)$.
- The S -matrix, depends only on the rapidity difference of the incoming/outgoing particles (Lorentz invariance) $\theta = \theta_a - \theta_b$.

Integrability and Scattering Theory

- The S -matrix theory of IQFTs has two key features:
 $\Rightarrow$ **factorisation** into two-body scattering processes



$\Rightarrow$ **no particle production**

- The full scattering theory is determined by **two-body scattering amplitudes** $S_{ab}^{cd}(\theta)$, where θ is the rapidity difference between the incoming (outgoing) particles.
- In many cases incoming and outgoing particles have exactly the same quantum numbers so we can write $S_{ab}(\theta)$ and we call these theories **diagonal**.
- A particle of type a has mass m_a , momentum $p_a(\theta) = m_a \sinh \theta$ and energy $e_a(\theta) = m_a \cosh \theta$.

Famous Examples

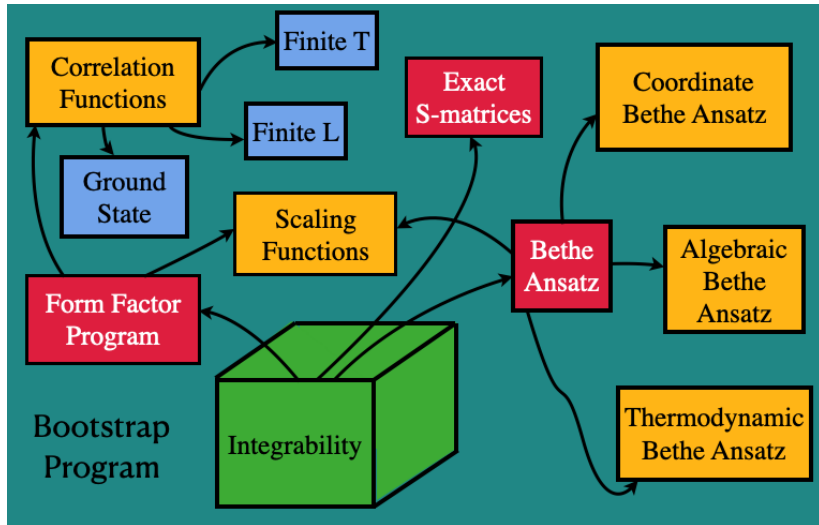
Let $[x]_\theta = \frac{\tanh \frac{1}{2}(\theta + i\pi x)}{\tanh \frac{1}{2}(\theta - i\pi x)}$ then:

- $S(\theta) = \pm 1$ for **free bosons/fermions**
- $S(\theta) = [-\frac{B}{2}]_\theta = [\frac{B}{2}]_\theta^{-1}$ for the **sinh-Gordon model**
- $S(\theta) = [\frac{1}{3}]_\theta$ for the **Lee-Yang model**
- $S_{ss}^{ss}(\theta) = S_{\bar{s}\bar{s}}^{\bar{s}\bar{s}}(\theta) = -\exp \left[-i \int_0^\infty \frac{dt}{t} \frac{\sinh \frac{\pi t(1-\xi)}{2} \sinh(t\theta)}{\sinh \frac{\pi t\xi}{2} \cosh \frac{\pi t}{2}} \right],$
 $S_{s\bar{s}}^{s\bar{s}}(\theta) = \frac{\sinh \frac{\theta}{\xi}}{\sinh \frac{i\pi-\theta}{\xi}} S_{ss}^{ss}(\theta)$ and $S_{\bar{s}s}^{\bar{s}s}(\theta) = \frac{\sinh \frac{i\pi}{\xi}}{\sinh \frac{i\pi-\theta}{\xi}} S_{ss}^{ss}(\theta),$
 $S_{sb_1}(\theta) = [\frac{1+\xi}{2}]_\theta, S_{b_1b_1}(\theta) = [\xi]_\theta, S_{b_2b_2}(\theta) = [\xi]_\theta^2 [2\xi]_\theta \cdots$
for the **sine-Gordon model**
- All the above multiplied by a CDD factor $e^{-im_a m_b \alpha \sinh \theta}$ describe the **T $\bar{T}$ -deformed versions** of the same theories.
- We have $[x]_\theta [x]_{-\theta} = 1$ and $[x]_\theta [x]_{i\pi-\theta}$ that is **unitarity** and **crossing** so that for $x < 0$ these are CDD factors too!

Some Applications

- A **wide range of physical phenomena** within integrability
- **Magnetism:** Heisenberg chain/Potts model
 - **Superconductivity:** Hubbard model
 - **Algebraic Structures:** Quantum groups, Yangians, Temperley-Lieb algebras, dynamical Yang-Baxter equation...
 - **Quantum Information:** Entanglement properties
 - **Hidden Structures in Higher Dimensions:** Planar $\mathcal{N} = 4$ Super Yang-Mills theory
 - **String Theory:** Operator Content, S -matrices in AdS_n theories
 - **Non-Isolated Systems:** Lax/Liouville/Liouvilian
 - **Generalised Gibbs Ensembles:** Thermalisation vs Equilibration
 - **Generalised Hydrodynamics:** out-of-equilibrium phenomena and experiments
 - **Cellular Automata:** Emergence of stable solutions, rule 54 ...
 - **Integrability in Higher Dimensions:** Tetrahedron Equation

Integrability as a Toolbox



Thermodynamic Bethe Ansatz: Motivation (I)

$$\varepsilon_a(\theta) = m_a \beta \cosh \theta - \sum_{b=1}^{\ell} \varphi_{ab} * L_b(\theta), \quad a = 1, \dots, \ell$$

- The scattering matrices of integrable quantum field theories (IQFTs) can be constructed as solutions to a set of consistency equation which **almost** entirely fix the S -matrix.
- There are certain uncertainties, such as **CDD factors**. How do we make sure that we have the correct S -matrix?
- There is strong motivation to develop **consistency checks** for integrable S -matrices: that is tools to extract information about the underlying CFT by employing just the S -matrix and particle spectrum as input.
- The TBA can be thought of as a consistency check, but it is much more than that.

Thermodynamic Bethe Ansatz: Motivation (II)

- The TBA provides a description of the **equilibrium** thermodynamics of IQFTs.
- It is written in the language of **quasiparticle excitations**, a description that is very natural, physical and useful in many applications.
- Allows us to compute average densities, RG flows and recover the underlying UV physics of the theory, all the while encoding interesting **mathematical structures**.
- Can be generalised to **non-equilibrium states** (GGEs) and even further when combined with **hydrodynamics** (GHD).

Bethe Wave Function

- Consider an IQFT and a set of N particles. If these particles are quite far away from each other, interactions are weak and the system can still be described by a **Bethe Wave Function** $\Psi(x_1, \dots, x_N)$ where x_i are particle positions.
- Let the system be compactified onto a circle of length L . Then, the Bethe wave function must be periodic:

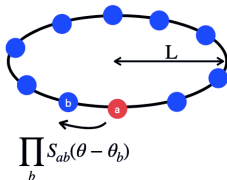
$$\Psi(x_1, \dots, x_a = 0, \dots, x_N) = \Psi(x_1, \dots, x_a = L, \dots, x_N)$$

- In other words, the wave function should not change if a particle takes **a trip around the world**.
- However, when it does so, it will come close and then scatter with all other particles, and non-trivial two-body interactions will kick in. It is here that **integrability** starts to play a crucial role!

Bethe Ansatz Equations

- Periodicity of the wave function is equivalent to:

$$e^{iLm_a \sinh \theta} \prod_{b \neq a} S_{ab}(\theta - \theta_b) = 1, \quad a = 1, \dots, N$$



- The logarithm of this equation (times $-i$) gives rise to the famous **Bethe ansatz equations**:

$$Lm_a \sinh \theta + \sum_{b \neq a} \delta_{ab}(\theta - \theta_b) = 2\pi n_a, \quad \delta_{ab}(\theta) = -i \log S_{ab}(\theta)$$

Roots and Holes

- Much can be said about the solutions to these equations. For each particle of type a there will be a set of **allowed values** of $n_a = \{n_a^1, \dots, n_a^{N_a}\}$ which are integers and possibly repeating for bosons and distinct for fermions.
- For each set of values there will be a corresponding set of solutions to the equations $\theta_a^{(i)}$ with $a = 1, \dots, N$ and $i = 1, \dots, N_a$. Each such solution is called a **root**.
- There are also sets of values $n_a^{(i)}$ that may be skipped. The solutions $\theta_a^{(i)}$ associated to “skipped” choices of $n_a^{(i)}$ are called **holes**.
- In the **thermodynamic or continuum limit** the densities of roots and holes will play an important role.

Root/Hole Densities in the Thermodynamic Limit

- Let $N_a, L \rightarrow \infty$ with N_a/L finite. This is the **thermodynamic limit**.
- In this limit we may define **root and hole densities**:

$$\rho_p^a(\theta) = \frac{1}{L} \frac{dn_a^{(r)}}{d\theta} = \text{number of roots with rapidities in } [\theta, \theta + d\theta]$$

$$\rho_h^a(\theta) = \frac{1}{L} \frac{dn_a^{(h)}}{d\theta} = \text{number of holes with rapidities in } [\theta, \theta + d\theta]$$

- The **total density of states per unit length** is:

$$\boxed{\rho_s^a(\theta) = \rho_p^a(\theta) + \rho_h^a(\theta) = \frac{1}{L} \frac{dn_a}{d\theta}}$$

Thermodynamic Limit of the BA Equations

- In the thermodynamic limit the θ -derivative of the Bethe ansatz equations becomes:

$$Lm_a \sinh \theta_a + \sum_{b \neq a} \delta_{ab}(\theta_a - \theta_b) = 2\pi n_a$$

$$\Downarrow \times \frac{1}{2\pi L} \frac{d}{d\theta_a}$$

$$\rho_s^a(\theta) = \frac{m_a}{2\pi} \cosh \theta + \sum_{b=1}^N (\varphi_{ab} * \rho_p^b)(\theta) \quad \text{with} \quad \varphi_{ab}(\theta) = \delta'_{ab}(\theta)$$

$$\text{and } (f * g)(\theta) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\beta f(\theta - \beta) g(\beta).$$

- These equations are still hard to solve but they give us a relationship between the state and root densities.
- It is convenient to define the **pseudoenergies** $\varepsilon_a(\theta)$ as

$$\frac{\rho_p^a(\theta)}{\rho_s^a(\theta)} = \frac{1}{e^{\varepsilon_a(\theta)} \pm 1}$$

for fermions (+) and bosons (-).

- In 1+1D IQFT particle statistics is generally determined by the value of $S_{aa}(0)$ which is -1 for most interesting examples.

Thermodynamic Equilibrium

- The equations above can be manipulated to generate equations for the pseudo-energies $\varepsilon_a(\theta)$.
- This is achieved by requiring thermodynamic equilibrium or the **minimisation of the free energy** per unit length.
- For instance, the total energy of the system is

$$h(\rho_p) = \sum_{a=1}^N m_a \int_{-\infty}^{\infty} d\theta \rho_p^a(\theta) \cosh \theta$$

- And the total thermodynamic entropy:

$$s(\rho_s, \rho_p) = \sum_{a=1}^N \int_{-\infty}^{\infty} d\theta (\mp \rho_s^a \log \rho_s^a - \rho_p^a \log \rho_p^a \pm (\rho_s^a \pm \rho_p^a) \log(\rho_s^a \pm \rho_p^a))$$

for bosons/fermions. This comes from **counting configurations** for fermions/bosons and then taking the continuum limit.

- The free energy is $f(\rho_s, \rho_p) = h(\rho_p) - T s(\rho_s, \rho_p)$.
Minimisation means $\frac{df}{d\rho_p} = 0$.

Thermodynamic Bethe Ansatz Equations

- The minimisation constraint combined with the relationship between ρ_s and ρ_p gives the **TBA equations**

Thermodynamic Bethe Ansatz Equations

$$\varepsilon_a(\theta) = m_a \beta \cosh \theta - \sum_{b=1}^N \varphi_{ab} * L_b(\theta), \quad a = 1, \dots, N, \quad \beta := \frac{1}{T}$$

$L_a(\theta) = \pm \log(1 \pm e^{-\varepsilon_a(\theta)})$ for fermions (+) and bosons (-).

- They are **coupled integral equations** for $\varepsilon_a(\theta)$.
- Even for complicated theories, they can be easily solved numerically through a recursive procedure.
- For free theories $\varphi_{ab}(\theta) = 0$ and $\varepsilon_a(\theta) = m_a \beta \cosh \theta$ are simply the free “on-shell” energies.
- The **extremal free energy** is simply

$$f(\beta) = -\frac{1}{\beta} \sum_{a=1}^N m_a \int_{-\infty}^{\infty} \frac{d\theta}{2\pi} \cosh \theta L_a(\theta)$$

Some References

Reviews/Books on Integrability:

- Patrick E. Dorey, Exact S -matrices [hep-th/9810026](#).
There are recorded lectures from 2015 easily found online.
- Giuseppe Mussardo, Statistical Field Theory: An Introduction to Exactly Solved Models in Statistical Physics, Oxford Graduate Texts, 2nd Edition (2020).
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- Ana L. Retore, Introduction to classical and quantum integrability, J. Phys. **A55** (2022) 173001; [2109.14280](#).

Reviews on Bethe Ansatz:

- Ludwig D. Faddeev, How Algebraic Bethe Ansatz Works for Integrable Models, [hep-th/9605187](#)
- Anastasia Doikou, Stefano Evangelisti, Giovanni Feverati and Nikos Karaiskos, Introduction to Quantum Integrability, [0912.3350](#)
- Fedor Levkovich-Maslyuk, Lectures on Bethe Ansatz, [1606.02950](#).